TWO-WAY DIGITAL CROSSOVER FOR MULTI-DRIVER AUDIO

Per-Band Equalization and Comparative Evaluation of
Complementary FIR and IIR Linkwitz–Riley (LR4) Crossover Designs

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Problem

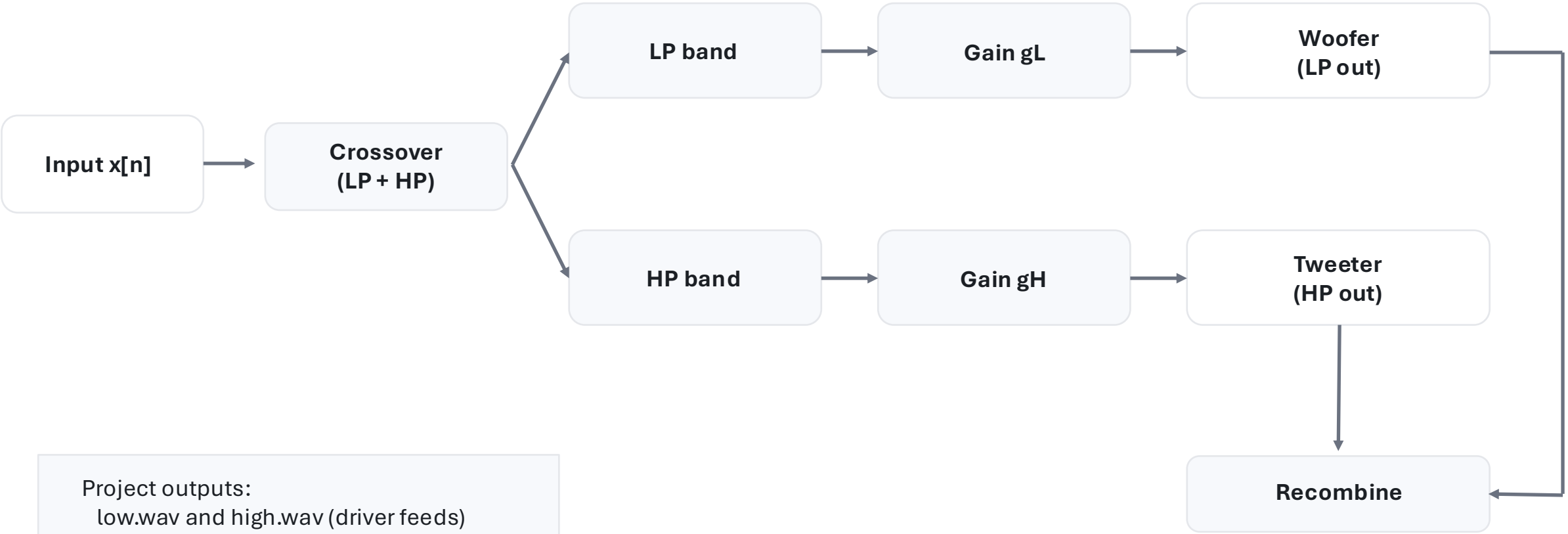
Why do audio products use multiple drivers?

- A single driver struggles to cover bass, mid, treble with low distortion.
- Different frequency ranges need different cone size, mass, and excursion.
- So products route low frequencies to a woofer and highs to a tweeter.
- After assembly, per-band tuning (EQ) is needed to balance the system.
- A crossover splits the spectrum and routes each band to the right driver

Example multi-way loudspeakers

Solution

Digital crossover + per-band EQ (active workflow)



Project outputs:
- low.wav and high.wav (driver feeds)
- sum.wav (for verification)
- sum_EQ.wav variants (EQ on/off demo)

Reproducibility

Reproducible pipeline

audio_in/

bass_pulse.wav
...

code/

run_all.m
apply_crossover.m
process_one_clip.m
design_iir_crossover_lr4.m
design_fir_crossover.m
test_iir_crossover_lr4.m
test_fir_crossover.m
make_paper_figures.m
cfg_default.m
recon_metrics.m

audio_out/

```
bass_pulse/
    FIR/
        sum.wav
        sum_EQ.wav
        low.wav
        low_EQ.wav
        high.wav
        high_EQ.wav
    IIR/
        sum.wav
        sum_EQ.wav
        low.wav
        low_EQ.wav
        high.wav
        high_EQ.wav
.../
```

Inputs → code → exported driver feeds

FIR vs IIR

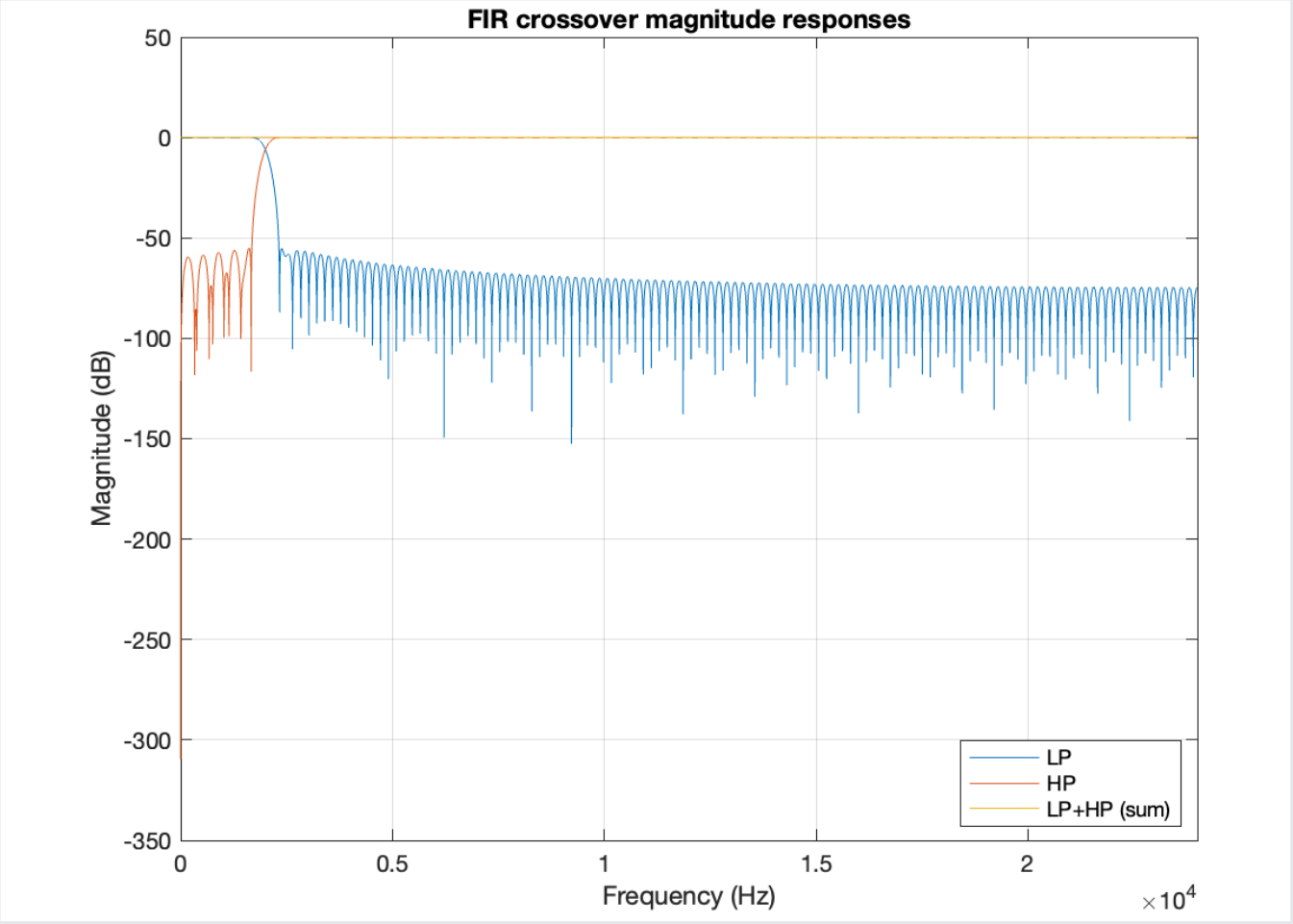
Two standard implementations (same crossover spec)

Aspect	FIR (complementary, linear-phase)	IIR (Linkwitz–Riley LR4)
Theoretical sum target	Pure delay $H_\Sigma = e^{-j\omega D}$	Unity magnitude (all-pass phase)
Expected magnitude behaviour	Flat	Flat
Expected phase / group delay	constant	Frequency-dependent
Expected time-domain outcome	Exact reconstruction after alignment	Possible waveform differences

Same spec: F_s = 48 kHz, f_c = 2 kHz

Verification

Frequency domain verification - FIR crossover



implementation:

The FIR crossover is implemented as a **complementary pair** of filters

A linear-phase low-pass FIR filter is designed at the target crossover frequency, and the high-pass branch is obtained by spectral complementarity rather than being designed independently.

The magnitude responses shown in the plot confirm the expected behaviour, with the low- and high-pass filters crossing at the crossover frequency and the summed magnitude remaining flat across the audio band

Key point
A flat magnitude sum is a necessary condition, not a sufficient one. Phase and group delay determine time-domain behaviour

FIR: complementary design;

Verification

Frequency domain verification - IIR crossover

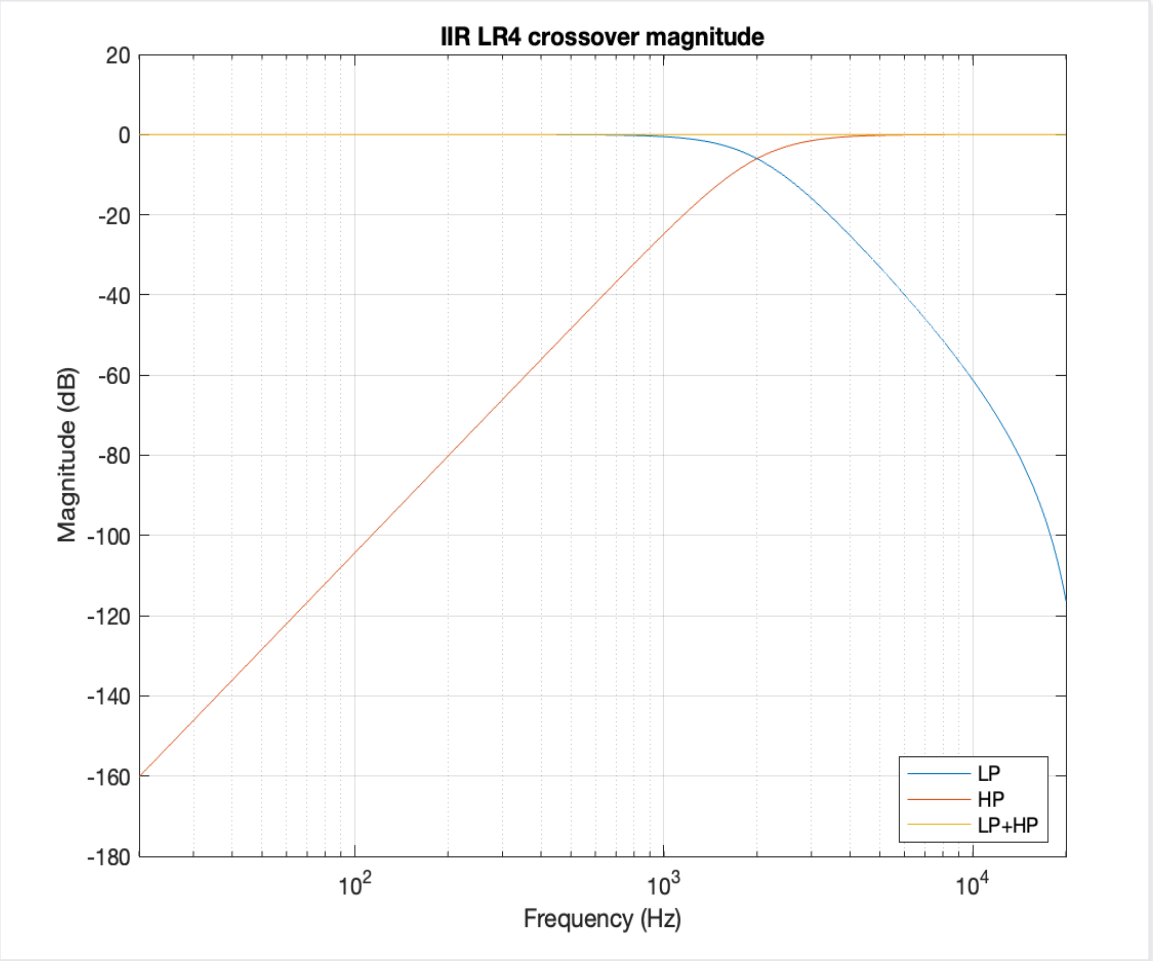
implementation:

The IIR crossover is implemented using a standard fourth-order Linkwitz–Riley (LR4) design. Each branch is obtained by cascading two second-order Butterworth filters with the same cutoff frequency, resulting in matched low-pass and high-pass responses.

The overall magnitude response should remain flat across the audio band. The plot confirms this expected behaviour: the low-pass and high-pass responses cross smoothly at the crossover frequency, and their sum remains unity in magnitude.

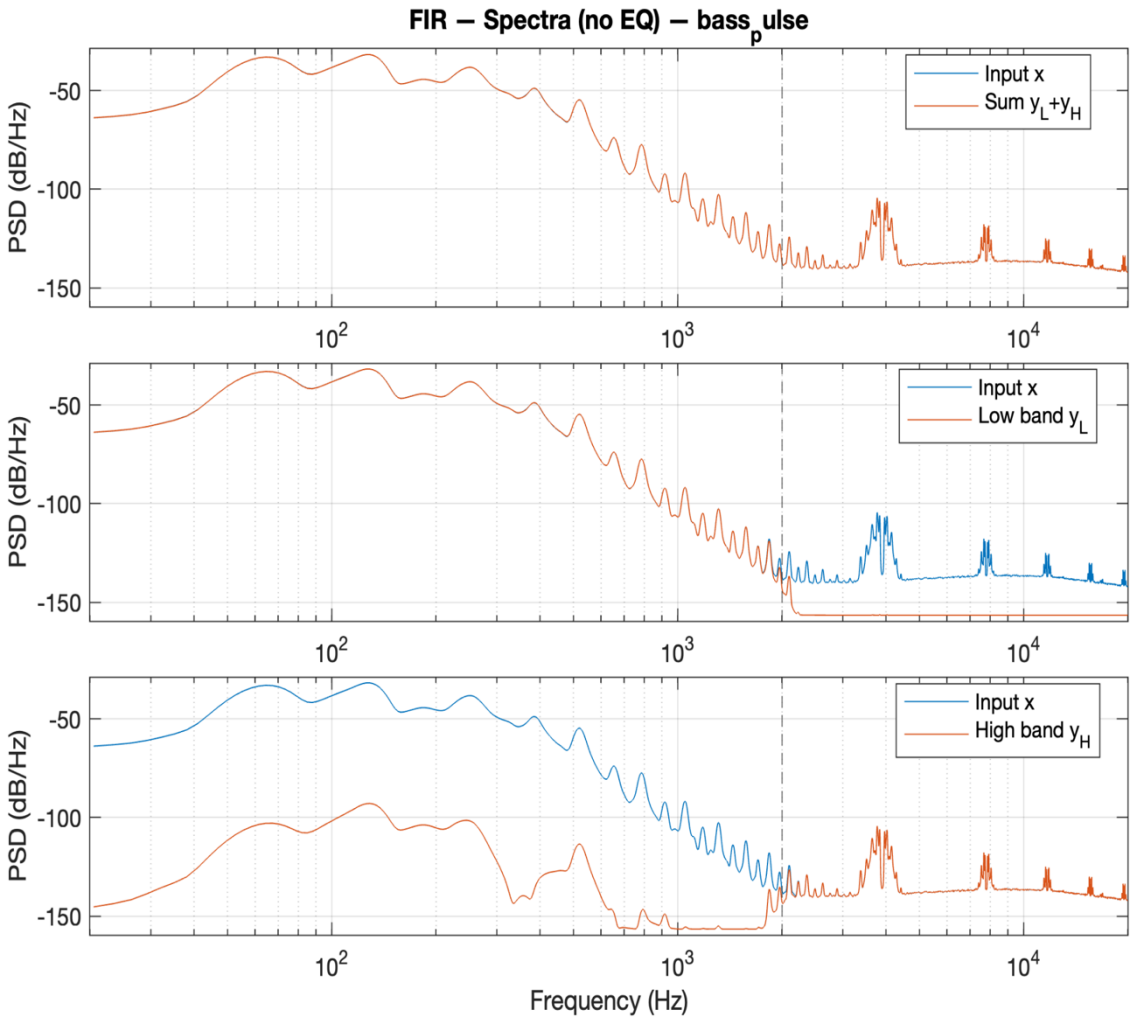
Unlike the FIR case, this flat sum is achieved without enforcing linear phase, meaning that the magnitude behaviour is correct by design, while phase and time-domain properties are left unconstrained and must be analysed separately

Key point
Magnitude-flat sum is guaranteed by design. Phase behaviour is not constrained and must be examined separately.

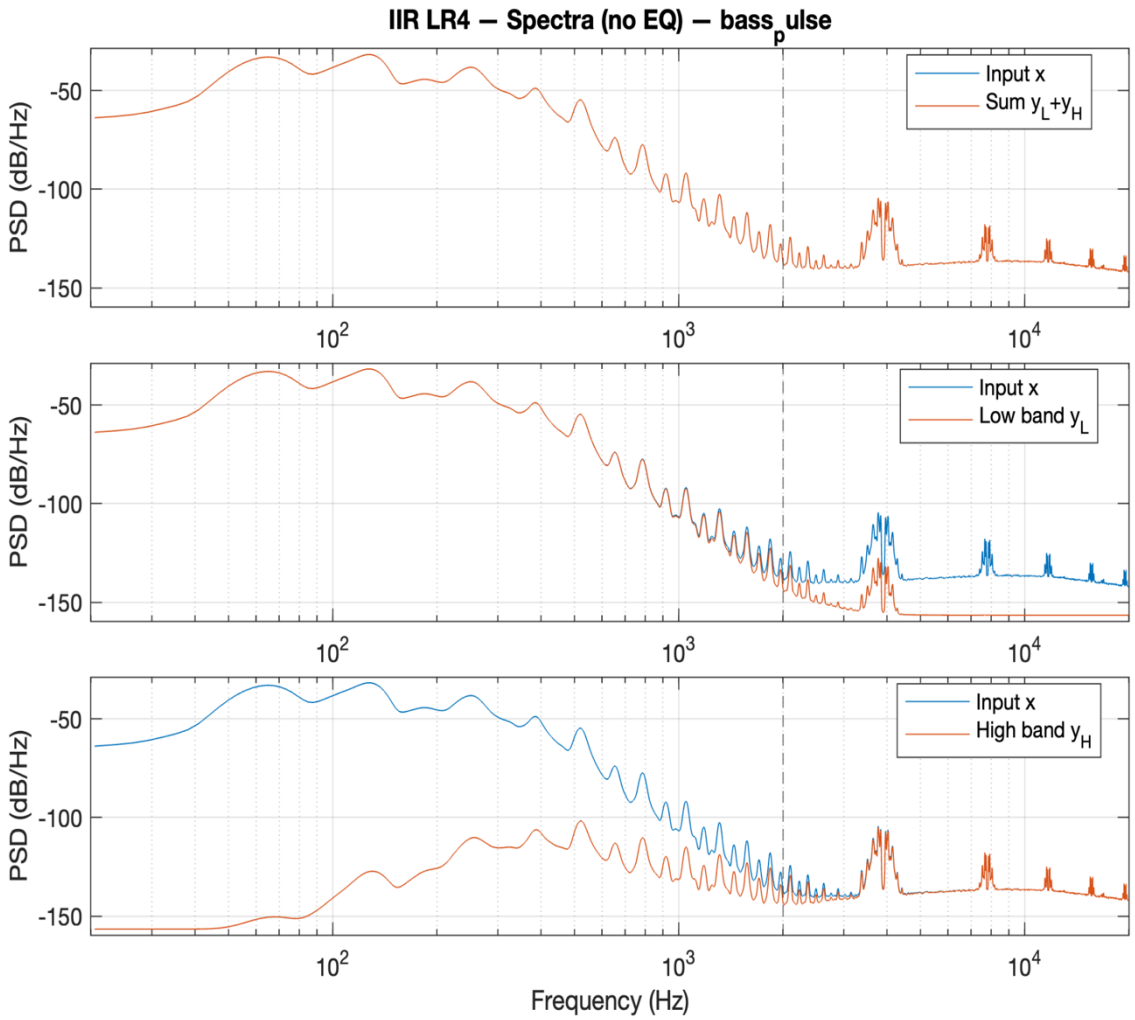


IIR: LR4 magnitude complementarity

Band splitting on a real clip (no EQ)

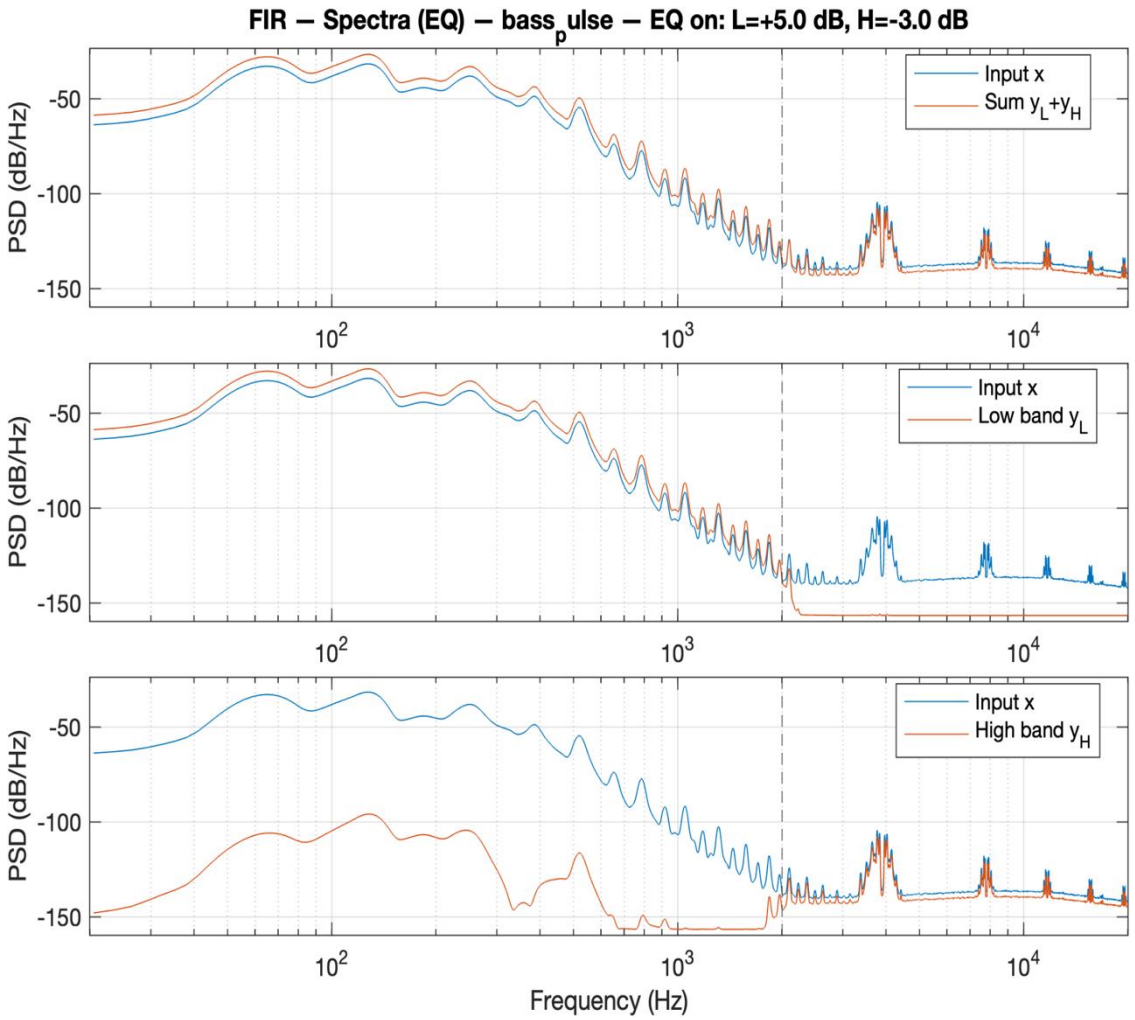


FIR spectra: input vs sum, low band, high band.

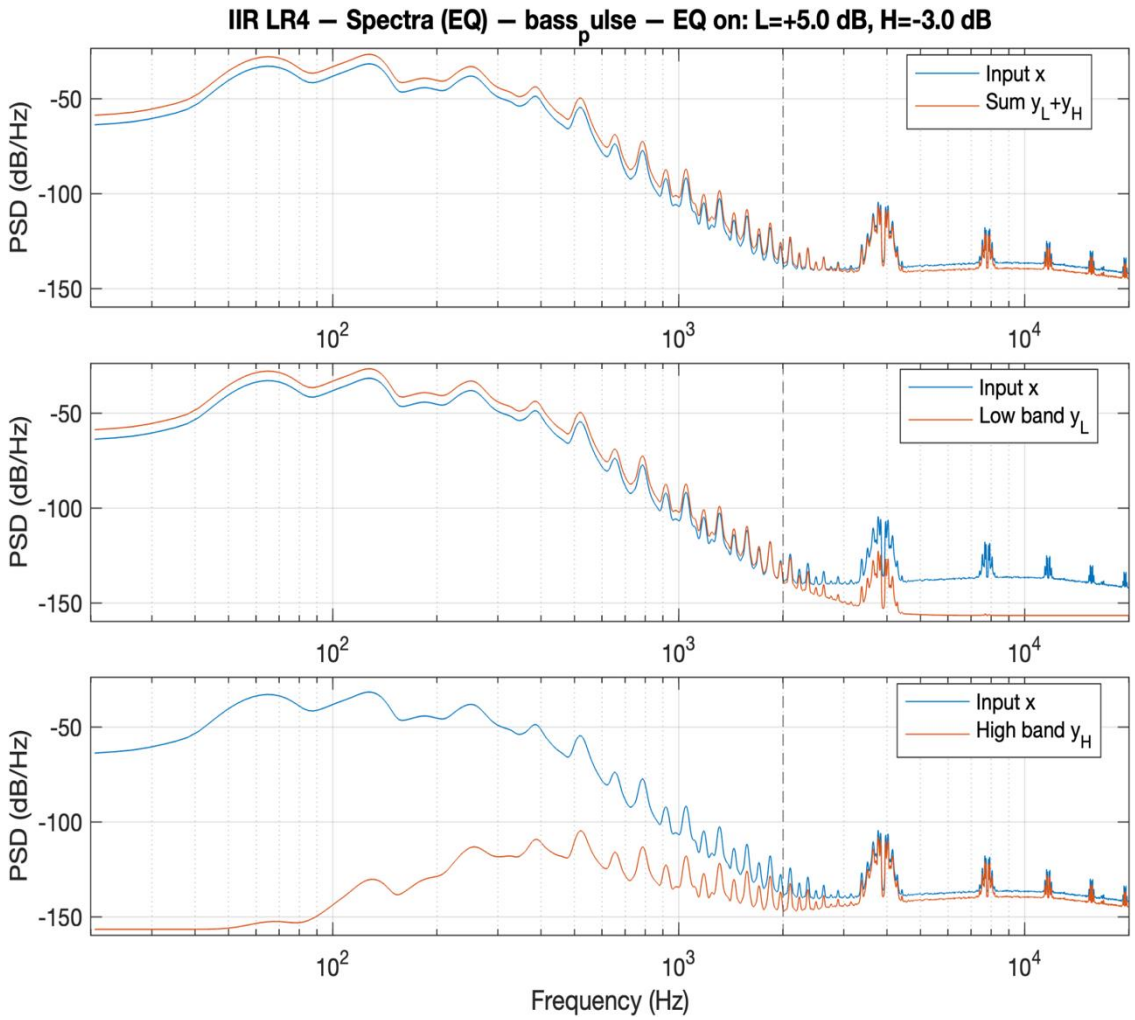


IIR spectra: input vs sum, low band, high band.

Per-band EQ: boosting bass and reducing highs

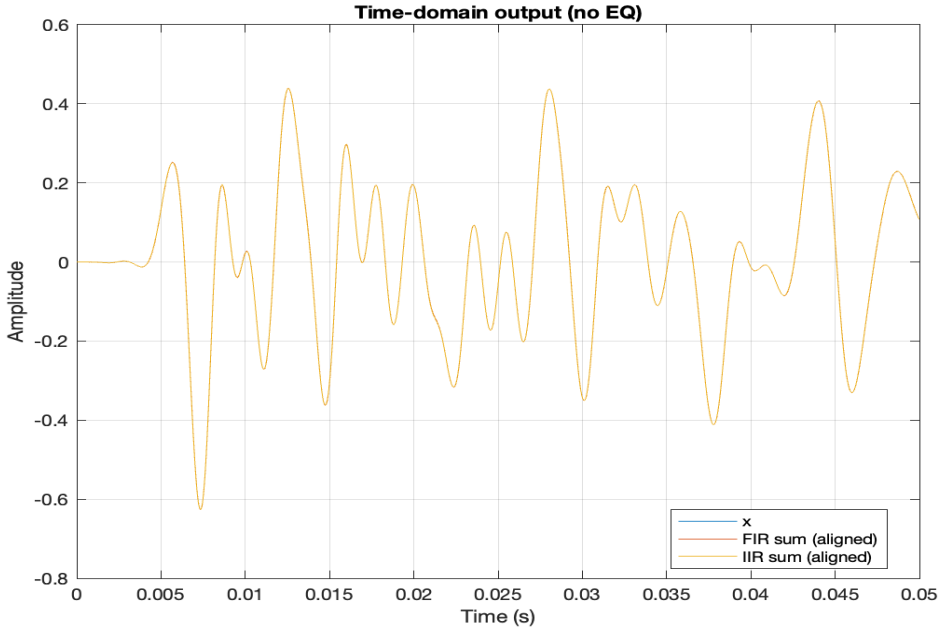


FIR with EQ (example: L = +5 dB, H = −3 dB).

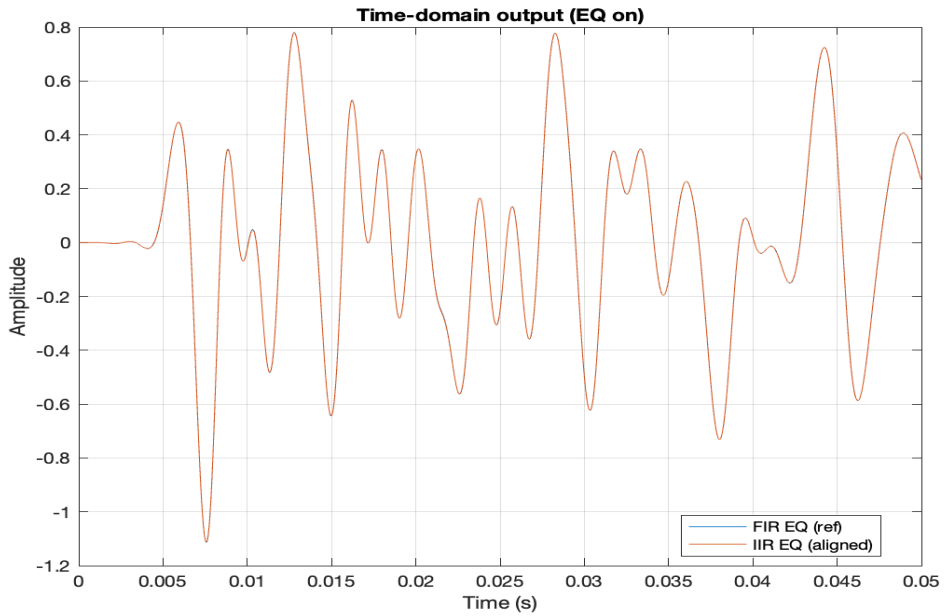


IIR with EQ (same settings).

Time-domain output



After alignment, the recombined output should match the input (or a delayed copy).



EQ changes the waveform because the spectral balance is modified.

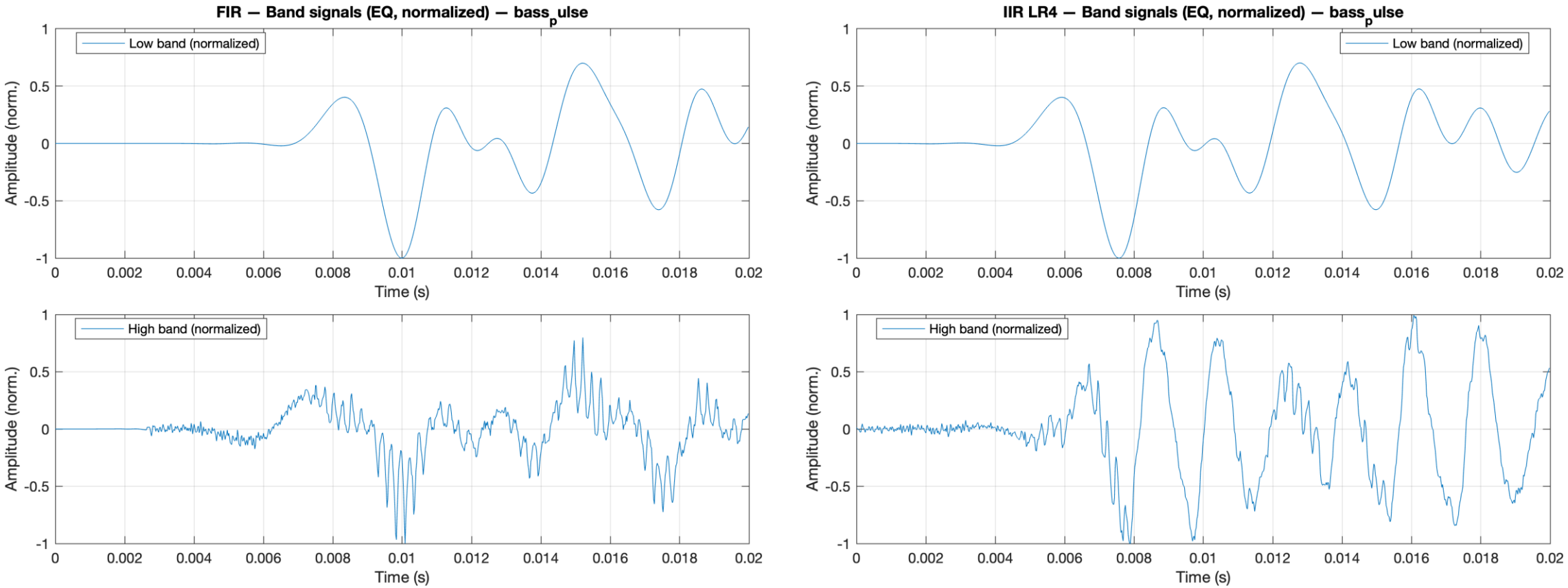
The time-domain plots show the recombined output signal before and after enabling per-band EQ.
In the no-EQ case, once the branches are aligned in time, the reconstructed signal closely matches the input waveform, up to a pure delay, confirming correct crossover recombination.
When EQ is applied, the waveform changes as expected: modifying the spectral balance of the signal alters its time-domain shape.

Band outputs - what each driver receives

Low- and high-frequency content are clearly separated and routed to dedicated outputs, confirming correct band isolation.

Differences between FIR and IIR are not in band allocation but in time-domain structure, consistent with their respective phase and group-delay behaviour.

These signals represent the actual inputs sent to the woofer and tweeter in an active loudspeaker system.



Low band → woofer; high band → tweeter. Both are exported as separate WAV files.

So what?

What this means in real systems

Choose FIR when...

- time-domain fidelity is need : the summed output is the input up to a pure delay (exact reconstruction after alignment).
- we want predictable phase / constant group delay, e.g., for transient-sensitive material or precise integration with other processing.
- we can afford latency and taps (more CPU/memory) to get linear-phase behaviour.

Choose IIR LR4 when...

- Low latency + efficiency is the priority (real-time / embedded / active speakers).
- we mainly care about a flat magnitude sum across the band and accept that phase is not a pure delay.
- we want a compact, standard crossover whose branches are matched around f_c (LR4 structure).
- our evaluation is perceptual / system-level, and small waveform differences are acceptable.

Choose based on constraints

Next

Conclusion

- A reproducible two-way digital crossover was designed and validated, exporting low-, high- and summed signals under a fixed, well-defined specification.
- FIR and IIR LR4 implementations satisfy the same magnitude-domain requirements but differ fundamentally in phase and time-domain behaviour.
- Complementary FIR filtering guarantees exact reconstruction up to a fixed delay, while IIR LR4 achieves magnitude-flat summation with nonlinear phase.
- Per-band EQ acts as a practical and controlled tuning stage, modifying spectral balance without breaking crossover complementarity.
- The choice between FIR and IIR is therefore driven by system constraints: time fidelity versus latency and computational efficiency.

Thanks!